

Introduction to Data Science — Final Exam Paper 4 (Medium)

Course Code: BCSE2C05 | Semester: II | Max Marks: 100 | Time: 3 Hours Difficulty: ★★☆☆☆ Medium

SECTION A — Short Answer (5 Marks Each, Do Any 4 of 5 = 20 Marks)

Q1. Explain Variance and Standard Deviation with a numerical example.

Variance (σ^2) measures how spread out data points are from the mean. **Standard Deviation (σ)** is the square root of variance — expressed in the same units as data.

Example: Data = {4, 6, 8, 10, 12}

Step 1: Mean = $(4+6+8+10+12)/5 = 40/5 = 8$

Step 2: Deviations from mean:

x_i	$x_i - \mu$	$(x_i - \mu)^2$
4	-4	16
6	-2	4
8	0	0
10	2	4
12	4	16

Step 3: Variance = $\Sigma(x_i - \mu)^2 / N = (16+4+0+4+16)/5 = 40/5 = 8.0$

Step 4: Standard Deviation = $\sqrt{8} = 2.83$

Q2. Differentiate between Data Integration and Data Transformation.

Aspect	Data Integration	Data Transformation
Definition	Combining data from multiple sources into a single unified dataset.	Converting data from one format/structure to another.
Purpose	Create a consistent, comprehensive dataset.	Make data suitable for analysis or modelling.
Challenges	Schema conflicts, naming inconsistencies, duplicate detection.	Choosing right scaling method, handling categorical data.
Techniques	Entity matching, schema alignment, ETL pipelines.	Normalization, encoding, aggregation, discretization.
Example	Merging hospital patient records from 3 different databases.	Converting salary from ₹ to normalized [0,1] range.

Q3. Write Python code to create a scatter plot and add a title, labels, and grid.

```
import matplotlib.pyplot as plt
import numpy as np

# Generate sample data
np.random.seed(42)
x = np.random.rand(50) * 100 # Study hours
y = x * 0.6 + np.random.randn(50) * 10 + 20 # Marks

# Create scatter plot
```

```
plt.figure(figsize=(8, 5))
plt.scatter(x, y, color='darkblue', marker='o', alpha=0.7, edgecolors='black')
plt.title('Study Hours vs Exam Marks', fontsize=14, fontweight='bold')
plt.xlabel('Study Hours', fontsize=12)
plt.ylabel('Marks Obtained', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

Q4. What is the difference between `loc` and `iloc` in Pandas? Give examples.

Feature	<code>loc</code>	<code>iloc</code>
Indexing	Label-based (uses row/column names).	Integer position-based (uses index numbers).
Syntax	<code>df.loc[row_label, col_label]</code>	<code>df.iloc[row_index, col_index]</code>
Inclusive	Both start and end are inclusive.	End is exclusive (like Python slicing).

```
import pandas as pd

df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Charlie', 'Diana'],
    'Age': [22, 25, 23, 28],
    'Marks': [85, 90, 78, 92]
}, index=['a', 'b', 'c', 'd'])

# loc - label-based
print(df.loc['a', 'Name'])      # Output: Alice
print(df.loc['a':'c', 'Name'])  # Output: Alice, Bob, Charlie (inclusive)

# iloc - position-based
print(df.iloc[0, 0])            # Output: Alice
print(df.iloc[0:3, 0])          # Output: Alice, Bob, Charlie (end exclusive)
```

Q5. What is Folium? Write code to display a map with a marker.

Folium is a Python library for creating interactive maps. It uses the Leaflet.js library and renders maps as HTML.

```
import folium

# Create a map centered on New Delhi
m = folium.Map(location=[28.6139, 77.2090], zoom_start=12)

# Add a marker
folium.Marker(
    location=[28.6139, 77.2090],
    popup='New Delhi, India',
    icon=folium.Icon(color='red', icon='info-sign')
).add_to(m)

# Save as HTML
m.save('delhi_map.html')
print("Map saved as delhi_map.html")
```

SECTION B — Analytical Questions (8 Marks Each, Do Any 5 of 6 = 40 Marks)

Q6. Explain the k-NN algorithm. How does the value of k affect model performance? Compute a classification manually.

k-NN (k-Nearest Neighbours): A supervised, lazy learning algorithm that classifies a new data point by looking at the majority class among its k nearest neighbours.

Effect of k:

- **k too small (e.g., 1):** Overfitting — sensitive to noise.
- **k too large (e.g., n):** Underfitting — model becomes too generalized.
- **Best practice:** Use odd k to avoid ties; find optimal k via cross-validation.

Manual Example:

Training data:

Point	X	Y	Class
A	1	2	Red
B	2	3	Red
C	5	5	Blue
D	6	7	Blue
E	3	3	Red

New point P = (4, 4), k = 3:

Euclidean distances:

- $d(P, A) = \sqrt{((4-1)^2 + (4-2)^2)} = \sqrt{(9+4)} = \sqrt{13} = \mathbf{3.61}$
- $d(P, B) = \sqrt{((4-2)^2 + (4-3)^2)} = \sqrt{(4+1)} = \sqrt{5} = \mathbf{2.24}$
- $d(P, C) = \sqrt{((4-5)^2 + (4-5)^2)} = \sqrt{(1+1)} = \sqrt{2} = \mathbf{1.41}$
- $d(P, D) = \sqrt{((4-6)^2 + (4-7)^2)} = \sqrt{(4+9)} = \sqrt{13} = \mathbf{3.61}$
- $d(P, E) = \sqrt{((4-3)^2 + (4-3)^2)} = \sqrt{(1+1)} = \sqrt{2} = \mathbf{1.41}$

3 nearest: C (Blue, 1.41), E (Red, 1.41), B (Red, 2.24)

Majority vote: 2 Red, 1 Blue → Predicted: Red

Q7. Given a dataset, write Pandas code to compute central tendency, handle missing data, and group by a column.

```
import pandas as pd
import numpy as np

# Create sample dataset
data = {
    'Department': ['CSE', 'ECE', 'CSE', 'ME', 'ECE', 'CSE', 'ME', 'ECE'],
    'Employee': ['Alice', 'Bob', 'Charlie', 'Diana', 'Eve', 'Frank', 'Grace', 'Henry'],
    'Salary': [55000, 60000, np.nan, 45000, np.nan, 70000, 48000, 52000],
    'Rating': [4.2, np.nan, 3.8, 4.5, 3.9, np.nan, 4.1, 4.3]
}
df = pd.DataFrame(data)

# Central tendency
print("Mean Salary:", df['Salary'].mean())
print("Median Salary:", df['Salary'].median())
print("Mode Department:", df['Department'].mode()[0])

# Missing values
```

```

print("\nMissing values:\n", df.isnull().sum())
df['Salary'] = df['Salary'].fillna(df['Salary'].median())
df['Rating'] = df['Rating'].fillna(df['Rating'].mean())
print("\nAfter imputation:\n", df.isnull().sum())

# Group by Department
dept_summary = df.groupby('Department').agg({
    'Salary': ['mean', 'max', 'min'],
    'Rating': 'mean'
}).round(2)
print("\nDepartment-wise Summary:")
print(dept_summary)

```

Q8. Explain Data Reduction. Compare PCA with Feature Selection.

Data Reduction reduces the volume of data while preserving key information for efficient analysis.

Why needed:

- Reduces computation time and memory.
- Avoids the curse of dimensionality.
- Improves model performance by removing redundant features.

Aspect	PCA (Principal Component Analysis)	Feature Selection
Approach	Creates new features (principal components) as linear combinations of original features.	Selects a subset of existing features.
Original Features	Transformed into new axes; originals are lost.	Original features are retained.
Interpretability	Low — components are mathematical constructs.	High — selected features are meaningful.
Information Loss	Minimal if enough components retained.	Possible if important features are dropped.
Techniques	Eigenvalue decomposition, SVD	Correlation analysis, Chi-square, Mutual Information
Use Case	High-dimensional data (images, genetics)	Tabular data with known feature meanings

Other Data Reduction Techniques:

- **Numerosity Reduction** — Represent data with models (regression coefficients) instead of raw values.
- **Data Compression** — Lossless (ZIP) or lossy (JPEG) encoding.
- **Sampling** — Random, systematic, or stratified sampling to reduce dataset size.

Q9. Write Python code to create a Boxplot and a Histogram using Seaborn and Matplotlib.

```

import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

# Sample data
np.random.seed(42)
data = {
    'Department': np.random.choice(['CSE', 'ECE', 'ME'], 100),
    'Salary': np.random.normal(50000, 15000, 100),
    'Performance': np.random.normal(75, 10, 100)
}

```

```

}
df = pd.DataFrame(data)

# — Boxplot (Seaborn) —
plt.figure(figsize=(10, 5))
sns.boxplot(x='Department', y='Salary', data=df, palette='pastel')
plt.title('Salary Distribution by Department', fontsize=14)
plt.xlabel('Department')
plt.ylabel('Salary (₹)')
plt.show()

# — Histogram (Matplotlib) —
plt.figure(figsize=(8, 5))
plt.hist(df['Performance'], bins=15, color='coral', edgecolor='black', alpha=0.7)
plt.title('Distribution of Performance Scores', fontsize=14)
plt.xlabel('Performance Score')
plt.ylabel('Frequency')
plt.axvline(df['Performance'].mean(), color='red', linestyle='--',
            label=f'Mean = {df["Performance"].mean():.1f}')
plt.legend()
plt.show()

```

Interpretation:

- **Boxplot** shows median (center line), Q1-Q3 (box), whiskers (1.5×IQR), and outliers (dots beyond whiskers).
- **Histogram** shows frequency distribution. The mean line helps identify skewness.

Q10. Explain the K-Means Clustering algorithm. Perform one iteration with given data.

K-Means is an unsupervised partitioning algorithm that groups data into K clusters.

Algorithm:

1. Choose K → Initialize K centroids
2. **Assign** each point to nearest centroid
3. **Update** centroids = mean of assigned points
4. **Repeat** until convergence

Given: Points = {1, 3, 7, 8, 9}, K = 2, C1 = 1, C2 = 7

Iteration 1 — Assignment:

Point	d to C1=1	d to C2=7	Cluster
1	0	6	C1
3	2	4	C1
7	6	0	C2
8	7	1	C2
9	8	2	C2

New centroids:

- C1 = (1+3)/2 = **2.0**
- C2 = (7+8+9)/3 = **8.0**

Iteration 2 — Assignment (C1=2.0, C2=8.0):

Point	d to C1=2	d to C2=8	Cluster
1	1	7	C1

3	1	5	C1
7	5	1	C2
8	6	0	C2
9	7	1	C2

Same clusters → **Converged!** Final: Cluster1 = {1, 3}, Cluster2 = {7, 8, 9}

Q11. What is Ethical Data Science? Discuss privacy, bias, and transparency with examples.

Ethical Data Science ensures that data collection, analysis, and deployment of models are conducted fairly, transparently, and without causing harm.

1. Data Privacy:

- Users must consent to data collection.
- Data must be stored securely (encryption, access controls).
- Regulations: GDPR (EU), IT Act (India), CCPA (California).
- **Example:** WhatsApp faced backlash in India (2021) for updating privacy policy to share data with Facebook without clear user consent.

2. Algorithmic Bias:

- Models can discriminate based on race, gender, or socioeconomic status if trained on biased data.
- **Example:** COMPAS algorithm (used in US courts) predicted Black defendants as higher recidivism risk compared to White defendants with similar profiles.
- **Mitigation:** Balanced training data, fairness-aware algorithms, regular audits.

3. Transparency:

- Users should understand how decisions affecting them are made.
- "Black box" models (deep neural networks) are hard to interpret.
- **Explainable AI (XAI)** techniques: LIME, SHAP values.
- **Example:** EU's GDPR gives citizens the "right to explanation" for automated decisions.

4. Informed Consent:

- **Example:** Cambridge Analytica (2018) — harvested 87M Facebook profiles without proper consent for political targeting.

SECTION C — Long Answer (20 Marks Each, Do Any 2 of 3 = 40 Marks)

Q12. (a) Explain Big Data, its 5 V's, and impact on industries (10 Marks). (b) Describe the Data Science Lifecycle with a real-world case study (10 Marks).

(a) Big Data & 5 V's:

Big Data = datasets too large and complex for traditional tools.

V	Meaning	Example
Volume	Terabytes to petabytes of data	YouTube: 500+ hours of video uploaded every minute
Velocity	Speed of generation & processing	Credit card transactions processed in milliseconds
Variety	Different data types	Text + images + GPS + audio from a single app
Veracity	Quality, accuracy, trustworthiness	Twitter data may contain bot-generated fake content
Value	Business insights after processing	Netflix saves \$1B/year through recommendation engine

Industry Impact:

1. **Healthcare** — Real-time patient monitoring via wearables.

- 2. **Banking** — Real-time fraud detection on millions of transactions.
- 3. **Retail** — Demand forecasting and inventory management.
- 4. **Smart Cities** — Traffic optimization using IoT sensor data.
- 5. **Agriculture** — Satellite + weather data for precision farming.

(b) Data Science Lifecycle — Zomato Case Study:

Phase	Application
Problem	"Predict restaurant delivery time to improve customer satisfaction."
Data Collection	Order history, restaurant location, driver GPS, traffic APIs, weather data.
Data Cleaning	Remove cancelled orders, fill missing delivery times, normalize addresses.
EDA	Visualize avg delivery time by city, day of week, cuisine type. Find peak hours.
Feature Engineering	Distance (restaurant to customer), time of order, restaurant rating, traffic index.
Modelling	Train Random Forest and XGBoost regression models.
Evaluation	Compare MAE and RMSE on test data. XGBoost: MAE = 4.2 mins.
Deployment	Real-time prediction displayed to customers: "Estimated delivery: 32 mins."

Q13. Write a complete Python program for employee data analysis using Pandas, NumPy, Matplotlib, and Seaborn.

File: sales_data.csv with columns: Sales_ID, Salesperson, Region, Product, Units_Sold, Revenue, Quarter, Year

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# — 1. Load & Inspect —
df = pd.read_csv('sales_data.csv')
print("Shape:", df.shape)
print("Data Types:\n", df.dtypes)
print("\nFirst 5 Rows:\n", df.head())
print("\nStatistical Summary:\n", df.describe())

# — 2. Central Tendency —
print(f"\nMean Revenue: ₹{df['Revenue'].mean():.2f}")
print(f"Median Revenue: ₹{df['Revenue'].median():.2f}")
print(f"Mode of Region: {df['Region'].mode()[0]}")
print(f"Std Dev of Units_Sold: {df['Units_Sold'].std():.2f}")

# — 3. Missing Values —
print("\nMissing Values:\n", df.isnull().sum())
df['Revenue'] = df['Revenue'].fillna(df['Revenue'].median())
df['Units_Sold'] = df['Units_Sold'].fillna(df['Units_Sold'].mean())
df['Region'] = df['Region'].fillna(df['Region'].mode()[0])

# — 4. Derived Column —
df['Revenue_Category'] = pd.cut(df['Revenue'],
                                bins=[0, 50000, 150000, float('inf')],
                                labels=['Low', 'Medium', 'High'])
print("\nRevenue Category Distribution:")
print(df['Revenue_Category'].value_counts())

# — 5. Group by Region —
```

```

region_summary = df.groupby('Region').agg({
    'Revenue': ['sum', 'mean'],
    'Units_Sold': 'sum'
}).round(2)
print("\nRegion-wise Summary:")
print(region_summary)

# — 6. Boxplot —
plt.figure(figsize=(10, 6))
sns.boxplot(x='Region', y='Revenue', data=df, palette='Set2')
plt.title('Revenue Distribution by Region')
plt.ylabel('Revenue (₹)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# — 7. Histogram —
plt.figure(figsize=(8, 5))
plt.hist(df['Units_Sold'], bins=20, color='steelblue', edgecolor='black')
plt.title('Distribution of Units Sold')
plt.xlabel('Units Sold')
plt.ylabel('Frequency')
plt.show()

# — 8. Heatmap (correlation) —
numeric_df = df.select_dtypes(include=[np.number])
plt.figure(figsize=(8, 6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Feature Correlation Heatmap')
plt.tight_layout()
plt.show()

# — 9. IQR Outlier Detection —
Q1 = df['Revenue'].quantile(0.25)
Q3 = df['Revenue'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[(df['Revenue'] < lower) | (df['Revenue'] > upper)]
print(f"\nOutlier Bounds: [{lower:,.0f}, {upper:,.0f}]")
print(f"Number of Outliers: {len(outliers)}")

# — 10. Save —
df.to_csv('sales_cleaned.csv', index=False)
print("\nSaved to sales_cleaned.csv")

```

Q14. (a) Compare Descriptive and Predictive Data Mining with case studies (8 Marks). (b) Explain sampling techniques with examples (12 Marks).

(a) Descriptive vs Predictive Mining:

Aspect	Descriptive	Predictive
Goal	Summarize past data	Forecast future outcomes
Question	"What happened?"	"What will happen?"
Techniques	Clustering, Association Rules	Classification, Regression
Output	Patterns, dashboards	Predictions, scores

Case Study — Banking:

- **Descriptive:** "65% of customers who defaulted had credit utilization > 80%." (pattern)
- **Predictive:** "Customer X has 72% probability of defaulting next month." (forecast)

Case Study — Retail:

- **Descriptive:** Market basket analysis → "Customers who buy bread also buy butter (support=0.7)."
- **Predictive:** "Customer segment A will spend ₹15,000 next quarter based on trend analysis."

(b) Sampling Techniques:

Technique	Method	Example	Pros	Cons
Simple Random	Every item has equal chance.	Drawing 100 names from a hat.	Unbiased.	May miss subgroups.
Systematic	Select every k-th item.	Every 10th product on assembly line.	Easy to implement.	Bias if data has cycles.
Stratified	Divide into strata, sample proportionally.	Sample students from each department proportionally.	All groups represented.	Need to know strata.
Cluster	Divide into clusters, randomly select entire clusters.	Select 5 random cities, survey all households in them.	Cost-effective for large areas.	High variability between clusters.
Convenience	Sample whatever is easily available.	Surveying friends on campus.	Quick and cheap.	Highly biased.

Numerical Example — Stratified Sampling:

- Population: 10,000 employees
- Strata: Engineering (4,000), Marketing (3,000), HR (2,000), Finance (1,000)
- Sample size: 500
- Proportional allocation: Eng=200, Mkt=150, HR=100, Fin=50